

The empirical recurrence for OEIS A301406, recovered from its tail

Adrian Perez Fontelles

Independent researcher

8 September 2026

Abstract

OEIS A301406 records “Empirical recurrence of order 45” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 126$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 48$. Its last coefficient is $c_{45} = -428616261304320 \neq 0$, so no further tail satisfies anything shorter; any order-45 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A301406 is “Number of $n \times 7$ 0..1 arrays with every element equal to 0, 1, 2, 3, 4 or 5 horizontally, vertically or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

64, 8192, 975492, 117605364, 14160945920, 1705287652128, 205353183605152, 24728921469216800, ...

The entry states:

Empirical recurrence of order 45 (see link above)

As of the “Last modified” line on the live entry (Mar 20 2018 11:11:01, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 126$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 126$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 45: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 45.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 252$ exact terms from $n = 3$ on, it returns order 45 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{45} c_i a(n-i),$$

c_1	85	c_{13}	1473665778505	c_{25}	−2452577773892032	c_{37}	313863630898593792
c_2	4074	c_{14}	3090629483394	c_{26}	−25141910472265792	c_{38}	33827237524406272
c_3	28475	c_{15}	−11328289112241	c_{27}	−27117047363531776	c_{39}	−105552908098994176
c_4	−607717	c_{16}	3840183385247	c_{28}	127957345346085888	c_{40}	−67290013338959872
c_5	−5775728	c_{17}	6392846507600	c_{29}	79651892511958272	c_{41}	−932575426117632
c_6	50002078	c_{18}	31788783332880	c_{30}	−340707612342744064	c_{42}	14019484943646720
c_7	380942138	c_{19}	−115276814062072	c_{31}	−149410537213594624	c_{43}	3656478531518464
c_8	−3032211312	c_{20}	239195867953876	c_{32}	518365854221332480	c_{44}	−713380109221888
c_9	−7860518186	c_{21}	250007730022672	c_{33}	295071682177875968	c_{45}	−428616261304320
c_{10}	102065075990	c_{22}	−2867154468314480	c_{34}	−495989544495611904		
c_{11}	−157444989376	c_{23}	3363593389433392	c_{35}	−389803833548406784		
c_{12}	−605891098941	c_{24}	5789539298466624	c_{36}	210737859422846976		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 48$, and it is the recurrence OEIS A301406 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{45} - c_1 x^{45-1} - \dots - c_{45}$. Its constant term is $-c_{45} = 428616261304320$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 45 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 45, so $\deg q = 45$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 45, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 12 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 45, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -428616261304320 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-45 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A301406.
- [2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)
- [4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)
- [5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.